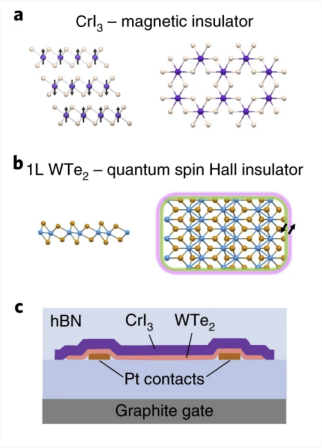
Proximity Induced Magnetism on Topological Edge States

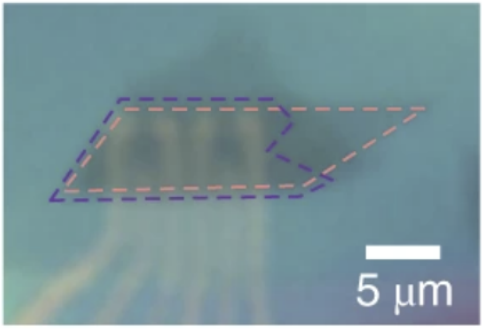
Master's Thesis Defense

László Erik Fritsch

June 16, 2026

Motivation





The trilayer CrI3 and monolayer WTe2 flakes are outlined by purple and pink dashed lines, respectively.

From Zhao, W., Fei, Z., Song, T. et al. Magnetic proximity and nonreciprocal current switching in a monolayer WTe2 helical edge. Nat. Mater. 19, 503–507 (2020)

Theoretical Framework: Multi-Scale Modeling

Real-Space Tight-Binding Hamiltonian:

$$\begin{aligned}\hat{H} = &\sum_{m_x=1}^{N_x}\sum_{m_y=1}^{N_y}\left(\hat{U}\otimes|m_x,m_y\rangle\langle m_x,m_y|\right)\\ &+\sum_{m_x=1}^{N_x-1}\sum_{m_y=1}^{N_y}\left(\hat{T}_x\otimes|m_x+1,m_y\rangle\langle m_x,m_y|+\text{h.c.}\right)\\ &+\sum_{m_x=1}^{N_x}\sum_{m_y=1}^{N_y-1}\left(\hat{T}_y\otimes|m_x,m_y+1\rangle\langle m_x,m_y|+\text{h.c.}\right)\end{aligned}$$

Extended Heisenberg Model:

$$H=\frac{1}{2}\sum_{i\neq j}J_{ij}^{\text{cl}}\vec{e}_i\cdot\vec{e}_j+\frac{1}{2}\sum_{i\neq j}\vec{D}_{ij}^{\text{cl}}\cdot(\vec{e}_i\times\vec{e}_j)+\frac{1}{2}\sum_{i\neq j}\vec{e}_i\cdot\bar{\bar{\Gamma}}_{ij}^{\text{cl}}\cdot\vec{e}_j+\sum_i\vec{e}_i\cdot K_i^{\text{cl}}\cdot\vec{e}_i$$

Generalized LKAG Torque Formalism:

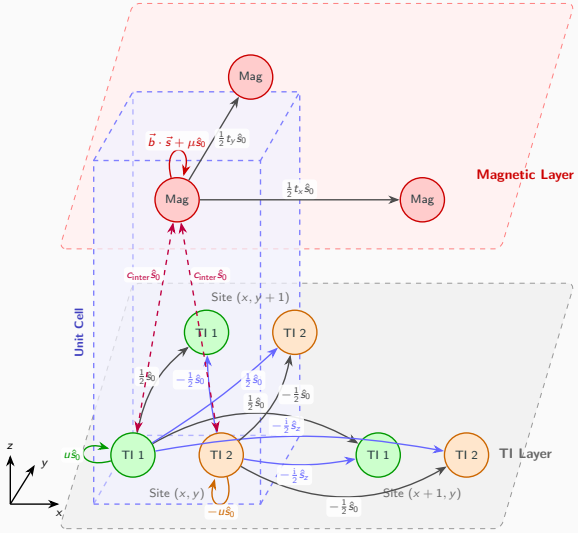
$$\delta E_{AB}^{int}=-\frac{1}{\pi}\int_{-\infty}^{E_F}\mathrm{d}\epsilon\,\mathrm{Im}\,\mathrm{Tr}\left[\hat{V}_A\hat{G}_o(\epsilon)\hat{V}_B\hat{G}_0(\epsilon)\right]$$

Tight-Binding Matrices & Real Space Model

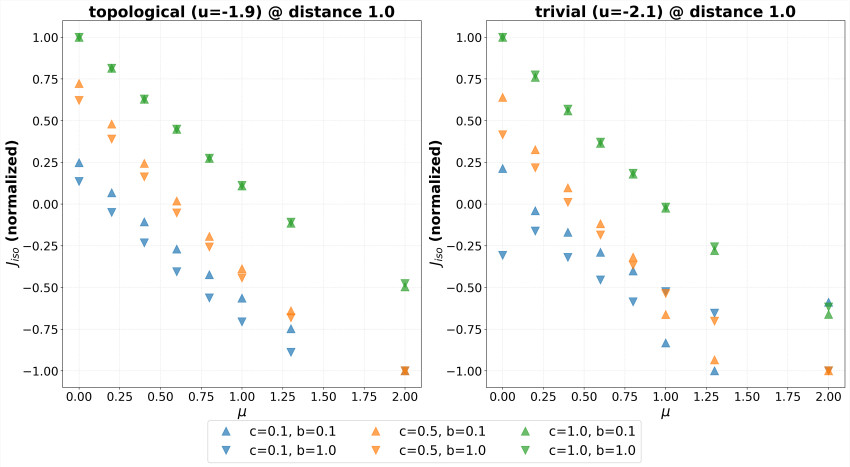
$$\hat{U} = \begin{bmatrix} u\hat{s}_0 & 0 & c_{\text{inter}}\hat{s}_0 \\ 0 & -u\hat{s}_0 & c_{\text{inter}}\hat{s}_0 \\ c_{\text{inter}}\hat{s}_0 & c_{\text{inter}}\hat{s}_0 & \vec{b}\cdot\vec{s} + \mu\hat{s}_0 \end{bmatrix}$$

$$\hat{T}_x = \frac{1}{2}\begin{bmatrix} \hat{s}_0 & -\mathrm{i}\hat{s}_z & 0 \\ -\mathrm{i}\hat{s}_z & -\hat{s}_0 & 0 \\ 0 & 0 & t_x\hat{s}_0 \end{bmatrix}$$

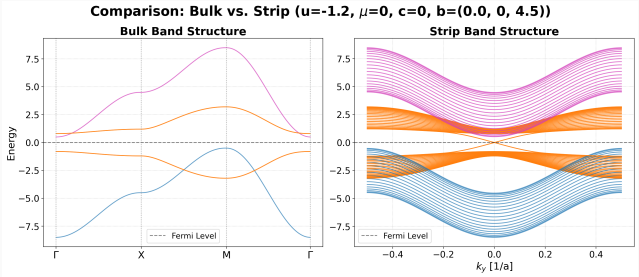
$$\hat{T}_y = \frac{1}{2}\begin{bmatrix} \hat{s}_0 & -\hat{s}_0 & 0 \\ \hat{s}_0 & -\hat{s}_0 & 0 \\ 0 & 0 & t_y\hat{s}_0 \end{bmatrix}$$

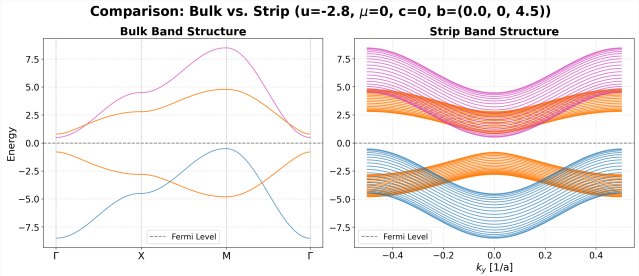


Parameter Space Mapping

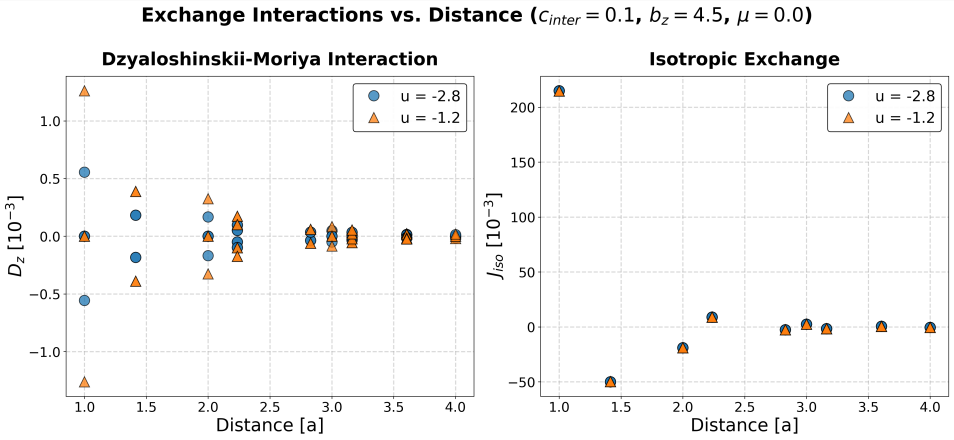


Band Structure

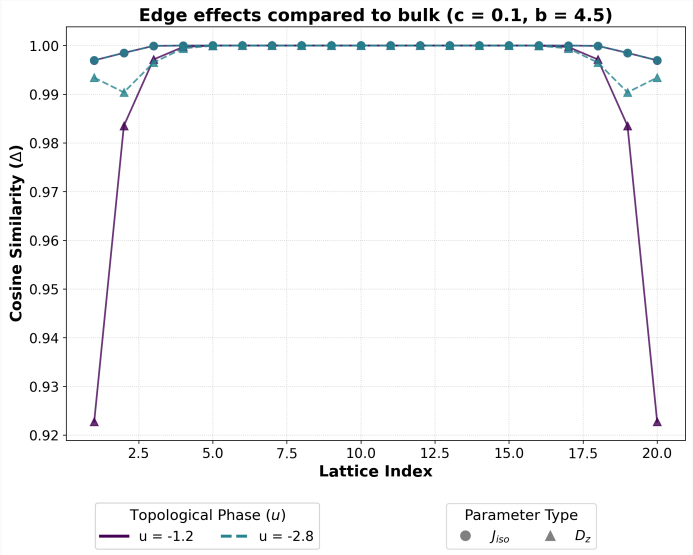




Exchange Interaction



Edge Effect


Edge effects compared to bulk (c = 0.1, b = 4.5)
Cosine Similarity (Δ)
1.00
0.99
0.98
0.97
0.96
0.95
0.94
0.93
0.92
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Lattice Index
Topological Phase (u)
u = -1.2
u = -2.8
Parameter Type
J_{iso}
D_z